

Git Assignment 1 Report

Summer Internship 2026

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Introduction to Git

Git is a widely used version control system that enables developers to organize and track changes made to project files. It supports collaborative development by allowing multiple contributors to work on the same project while preserving the complete history of updates. Known for its speed, stability, and flexibility, Git plays a significant role in modern software development. This assignment provided practical experience in using Git and GitHub to perform basic version control tasks.

Why Git is Used

Git is preferred in software development because of its powerful version control capabilities. Some of its key advantages include:

- Tracks every change made to project files.
 - Allows multiple developers to collaborate efficiently.
 - Supports branching and merging for parallel development.
 - Stores projects securely using remote repositories.
 - Makes project maintenance and code tracking easier.
 - Improves the overall software development workflow.
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Git Installation Process

The first step of the assignment involved setting up Git on the system to enable version control. The latest Git installer was obtained from the official website and

installed by following the standard setup procedure. After the installation was complete, Git Bash was launched to ensure that the software was functioning properly.

The command shown below was executed to check the installed version:

```
git --version
```

The command returned the current Git version, indicating that the installation was completed successfully and the environment was ready for use.

```
vishu@Vishnuvardhan MINGW64 ~  
$ git --version  
git version 2.54.0.windows.1
```

Git Configuration

Before working with the repository, Git was configured with the user's identity information. This ensures that every commit is associated with the correct author. The following commands were executed to set the username and email address:

```
git config --global user.name "Vishnu Vardhan"  
git config --global user.email "vtu30129@veltech.edu.in"
```

The configuration details were then checked using:

```
git config --global --list
```

The displayed information verified that the Git configuration was applied correctly.

```
vishu@Vishnuvardhan MINGW64 ~  
$ git config --global user.name "Vishnu vardhan"  
  
vishu@Vishnuvardhan MINGW64 ~  
$ git config --global user.email "vtu30129@veltech.edu.in"  
  
vishu@Vishnuvardhan MINGW64 ~  
$ git config --global --list  
user.name=Vishnu vardhan  
user.email=vtu30129@veltech.edu.in
```

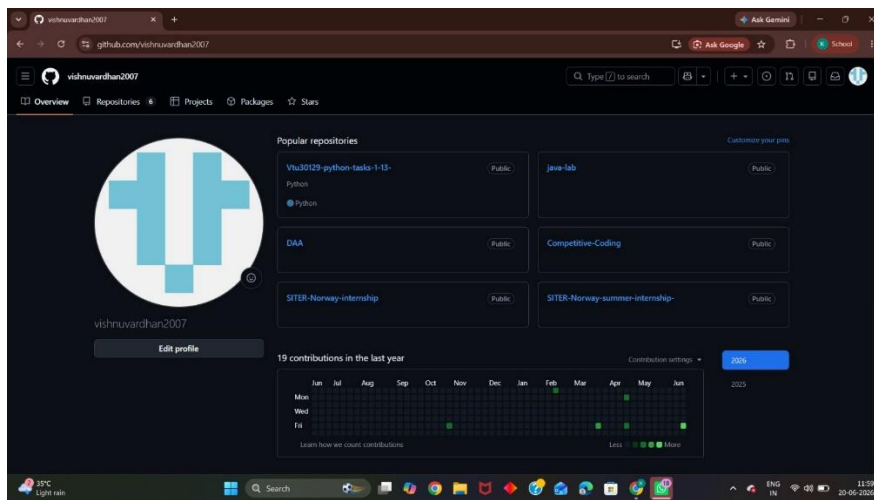
GitHub Profile Creation

A GitHub account was created to host the project repository online and manage the project using cloud-based version control. GitHub made it possible to store the project securely, access it from different devices, and keep the repository up to date.

GitHub Profile:

<https://github.com/vishnuvardhan2007>

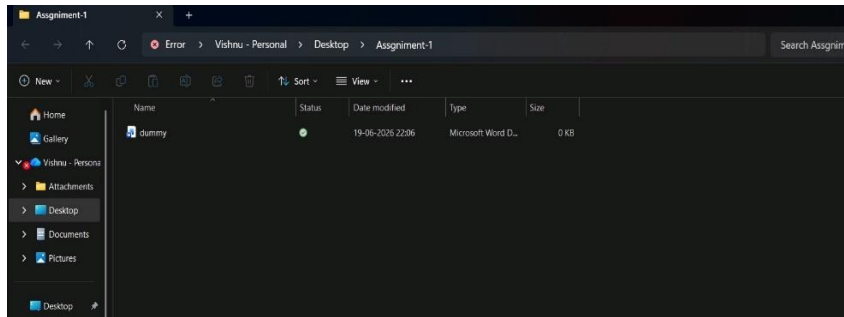
A separate repository was created for the internship assignment, and the project files were uploaded after each major update. This ensured proper version tracking and safe storage of the project.



Dummy Project Description

To gain practical experience with Git and GitHub, a simple project was created as part of this assignment. The project consisted of a document file named **"Dummy.docx"**, which served as the sample file for demonstrating version control.

The project was used to practice key Git operations, including file tracking, creating commits, and publishing the repository to GitHub. This exercise helped in understanding the complete Git workflow.



Git Commands Used

Various Git commands were executed to perform version control operations during the assignment:

- **Repository Creation:**

```
git init
```

- **Add Files to Staging Area:**

```
git add .
```

- **Record Changes:**

```
git commit -m "Initial Commit"
```

- **Configure Remote Repository:**

```
git remote add origin <repository-url>
```

- **Switch to Main Branch:**

```
git branch -M main
```

- **Publish Project to GitHub:**

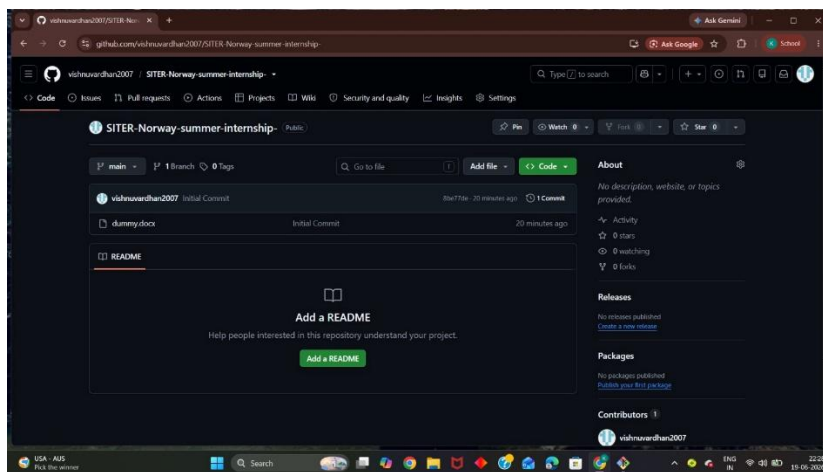
```
git push -u origin main
```

- **Display Commit Log:**

The above commands were essential for managing the repository, recording project updates, and synchronizing the local project with GitHub

Implementation of Git

This implementation provided practical experience in using Git for version control and demonstrated how GitHub can be used to store, manage, and share project repositories efficiently.



Challenges Faced

At the beginning of the assignment, understanding the Git workflow and remembering the correct sequence of commands was slightly challenging. Configuring Git, creating the repository, and connecting it to GitHub required a few attempts before the process became clear. With continuous practice, these issues were resolved successfully.

The experience improved my confidence in using Git and strengthened my understanding of version control concepts.

Conclusion

This assignment provided hands-on experience with Git and GitHub by covering the complete version control process. From installing and configuring Git to creating repositories, committing changes, and publishing the project on GitHub, every step contributed to a better understanding of repository management. Overall, the assignment highlighted the importance of version control in modern software development and collaborative projects.